1 **Design & Implementation of Human-Computer Interfaces**
NPTEL-MOOCS

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2 **L10 – Case Study (Usability Requirements to SRS)**

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4 **Recap**

- We are discussing the "requirement gathering, analysis & specification" stage – with a case study
  - So far discussed usability requirement gathering and specification of functional requirements
- Today – how to convert usability to functional specification (SRS)

5 **Example Scenario**

- A calendar app
  - We are interested to build a calendar app. It is meant for the students (mainly) to help in their various academic activities

6 **Usability Requirements**

7 **Calendar App – Usability Requirement Gathering**

- We performed contextual inquiry (CI)

8 **CI Recap - Five Stages**

- Plan – plan the process
- Initiate – start the process
- Execute – perform the process
- Close – end the process
- Reflect – analyze data

9 **Data Analysis**

- We followed affinity diagram method

10 **Affinity Diagram Method (Recap)**

- FIVE steps
  1. Generate Idea
  2. Display ideas
  3. Sort ideas into groups

4. Create "group header"
5. Draw finished diagram

11 **Assumption**
- We performed CI
  - With few students - they used paper-based, desktop and mobile calendars for academic activities only
  - Observations made and noted down (sticky-notes)
  - Affinity diagram created for analysis

12 **Affinity Diagram**

13 **How to Use?**

14 **Different Ways**
- Design guidelines/recommendations/principles
- "Creation" of functional requirements

15 **Guidelines**
- We can frame some guidelines based on observations

16 **Observations & Guidelines**

17 **Observations & Guidelines**

18 **Creation of FR**
- We can also add "new" functional requirements based on the observations

19 **FR Hierarchy**

20 **New FR (Example)**

R5: Views
Input: Date
Output: Specific view (timetable/submission deadlines/meeting list)
Description: In this function, the user can check day-wise timetable of lectures/upcoming assignment submission deadlines/scheduled meetings

21 **New FR (Example)**

R5.1: Timetable view
Input: Date
Output: Timetable for the day
Description: In this function, the user can check day-wise timetable of lectures

22 **New FR (Example)**

R5.2: Deadline view
Input: Date
Output: Submission deadlines
Description: In this function, the user can check day-wise assignment submission deadlines

23 **New FR (Example)**

R5.3: Meeting schedule

Input: Date
Output: List of meetings scheduled on the day
Description: In this function, the user can check day-wise meeting schedule

24 **FR Hierarchy (Revised)**

25 **Note**

- Similarly, we can add more such FRs (left as an exercise)